4-Two Elements sum to x

Problem Statement:

Given a sorted array of integers say arr[] and a number x. Write a recursive program using divide and conquer strategy to check if there exist two elements in the array whose sum = x. If there exist such two elements then return the numbers, otherwise print as “No”.

Note: Write a Divide and Conquer Solution

Input Format

First Line Contains Integer n – Size of array

Next n lines Contains n numbers – Elements of an array

Last Line Contains Integer x – Sum Value

Output Format

First Line Contains Integer – Element1

Second Line Contains Integer – Element2 (Element 1 and Elements 2 together sums to value “x”)

Answer:(penalty regime: 0 %)

```
#include <stdio.h>


int findPair(int arr[], int left, int right, int x)
{
  if (left >= right)
    return 0;

  int sum = arr[left] + arr[right];

  if (sum == x)
  {
    printf("%d\n", arr[left]);
    printf("%d\n", arr[right]);
    return 1;
  }

  if (sum < x)
    return findPair(arr, left + 1, right, x);
  else
    return findPair(arr, left, right - 1, x);
}

int main()
```

```
{
    int n, x;

    scanf("%d", &n);

    int arr[n];

    for (int i = 0; i < n; i++)
        scanf("%d", &arr[i]);

    scanf("%d", &x);

    if (!findPair(arr, 0, n - 1, x))
        printf("No\n");

    return 0;
}
```

Feedback

	Input	Expected	Got	
	4 2 4 8 10 14	4 10	4 10	
	5	No	No	

	Input	Expected	Got	
	2 4 6 8 10 100			

Passed all tests!